

## Lab 3: Docker Compose Multi-Container App

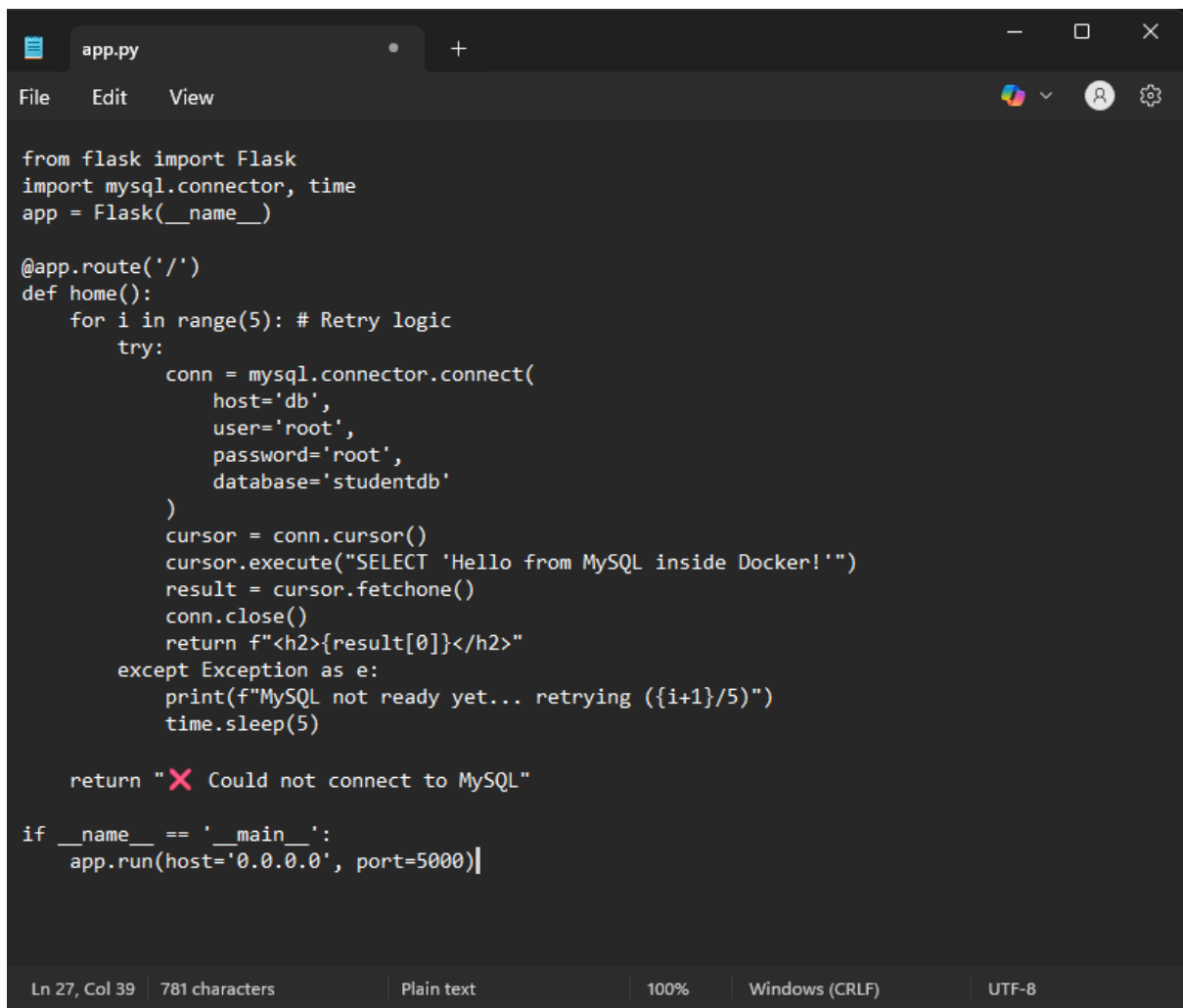
**Zunaira Abdul Aziz**

**BSSE23058**

### **Objectives:**

The main goal is to use Docker Compose to run an application with two containers. We will create a Python web app and a MySQL database that work together. We'll write a single docker-compose.yml file to connect and run both. This lab also teaches us how to use environment variables and save database data.

**Screenshot #1: code open in editor.**



The screenshot shows a code editor window with a dark theme. The title bar at the top indicates the file is named 'app.py'. The menu bar includes 'File', 'Edit', and 'View'. On the right side of the menu bar, there are icons for a color palette, a dropdown menu, a user profile, and settings. The code is written in Python and uses the Flask framework to connect to a MySQL database. It implements a retry logic for the database connection. The code is as follows:

```
from flask import Flask
import mysql.connector, time
app = Flask(__name__)

@app.route('/')
def home():
    for i in range(5): # Retry logic
        try:
            conn = mysql.connector.connect(
                host='db',
                user='root',
                password='root',
                database='studentdb'
            )
            cursor = conn.cursor()
            cursor.execute("SELECT 'Hello from MySQL inside Docker!'")
            result = cursor.fetchone()
            conn.close()
            return f"<h2>{result[0]}</h2>"
        except Exception as e:
            print(f"MySQL not ready yet... retrying ({i+1}/5)")
            time.sleep(5)

    return "❌ Could not connect to MySQL"

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000)
```

The status bar at the bottom of the editor shows the following information: 'Ln 27, Col 39', '781 characters', 'Plain text', '100%', 'Windows (CRLF)', and 'UTF-8'.

**Screenshot #2: Dockerfile content.**

```
zuni_2004@DESKTOP-16K3I1B x + v
zuni_2004@DESKTOP-16K3I1B:~/flask_mysql_app$ ls -l
total 4
drwxr-xr-x 2 zuni_2004 zuni_2004 4096 Nov  8 12:24 app
zuni_2004@DESKTOP-16K3I1B:~/flask_mysql_app$ ls
app
zuni_2004@DESKTOP-16K3I1B:~/flask_mysql_app$ cd app
zuni_2004@DESKTOP-16K3I1B:~/flask_mysql_app/app$ notepad.exe app.py
zuni_2004@DESKTOP-16K3I1B:~/flask_mysql_app/app$ notepad.exe requirements.txt
zuni_2004@DESKTOP-16K3I1B:~/flask_mysql_app/app$ notepad.exe Dockerfile
```

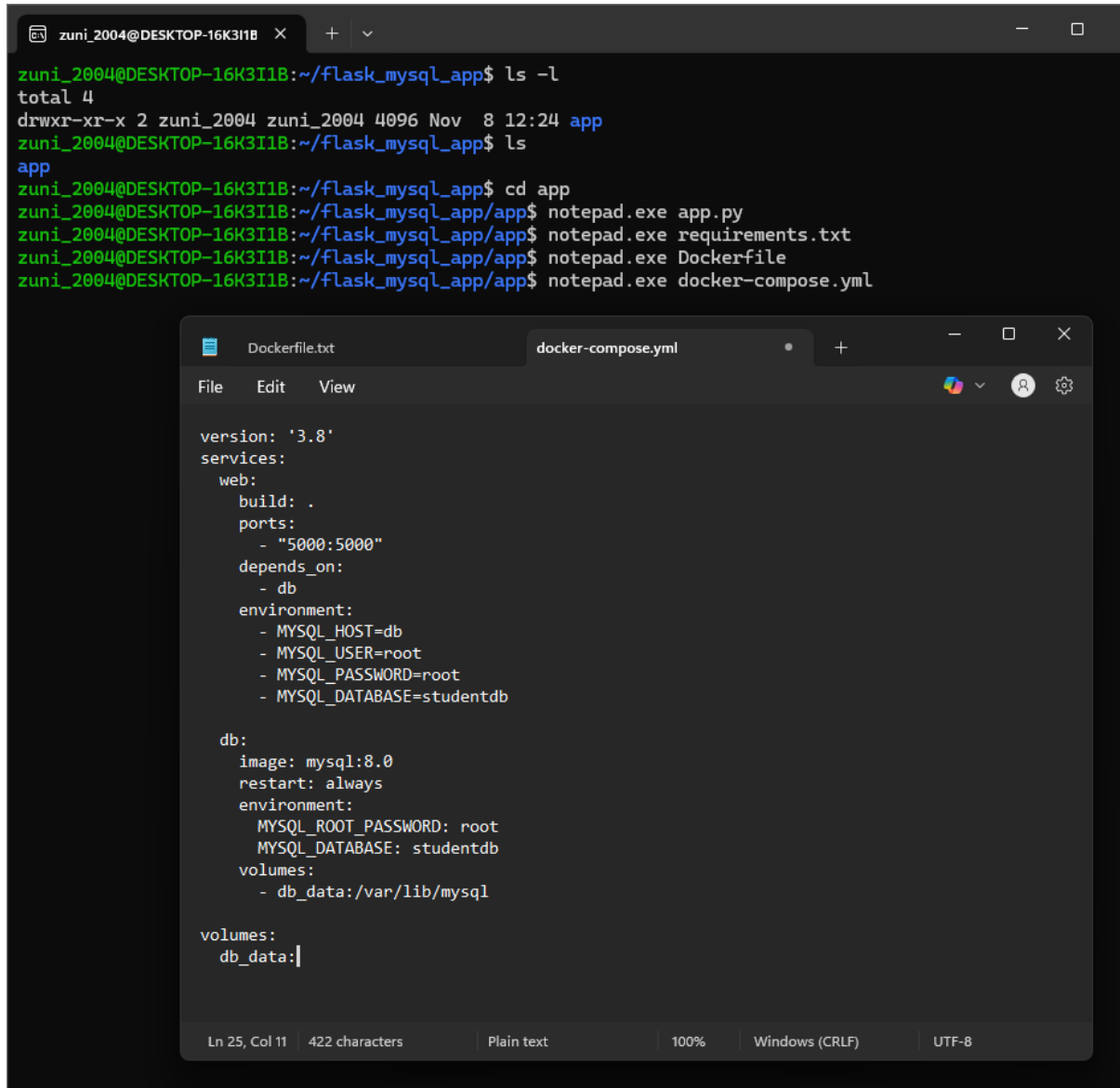
Dockerfile.txt +

File Edit View H1 ≡ B I ↺ Aa

FROM python:3.10-slim  
WORKDIR /app  
COPY app/ /app  
COPY requirements.txt .  
RUN pip install -r requirements.txt  
CMD ["python", "app.py"]

Ln 6, Col 23 134 characters Plain text 100% Windows (CRLF) UTF-8

**Screenshot #3: YAML file content.**



The screenshot shows a terminal window with the following commands and output:

```
zuni_2004@DESKTOP-16K3I1B:~/flask_mysql_app$ ls -l
total 4
drwxr-xr-x 2 zuni_2004 zuni_2004 4096 Nov  8 12:24 app
zuni_2004@DESKTOP-16K3I1B:~/flask_mysql_app$ ls
app
zuni_2004@DESKTOP-16K3I1B:~/flask_mysql_app$ cd app
zuni_2004@DESKTOP-16K3I1B:~/flask_mysql_app/app$ notepad.exe app.py
zuni_2004@DESKTOP-16K3I1B:~/flask_mysql_app/app$ notepad.exe requirements.txt
zuni_2004@DESKTOP-16K3I1B:~/flask_mysql_app/app$ notepad.exe Dockerfile
zuni_2004@DESKTOP-16K3I1B:~/flask_mysql_app/app$ notepad.exe docker-compose.yml
```

Below the terminal, a Notepad window titled 'docker-compose.yml' displays the following YAML content:

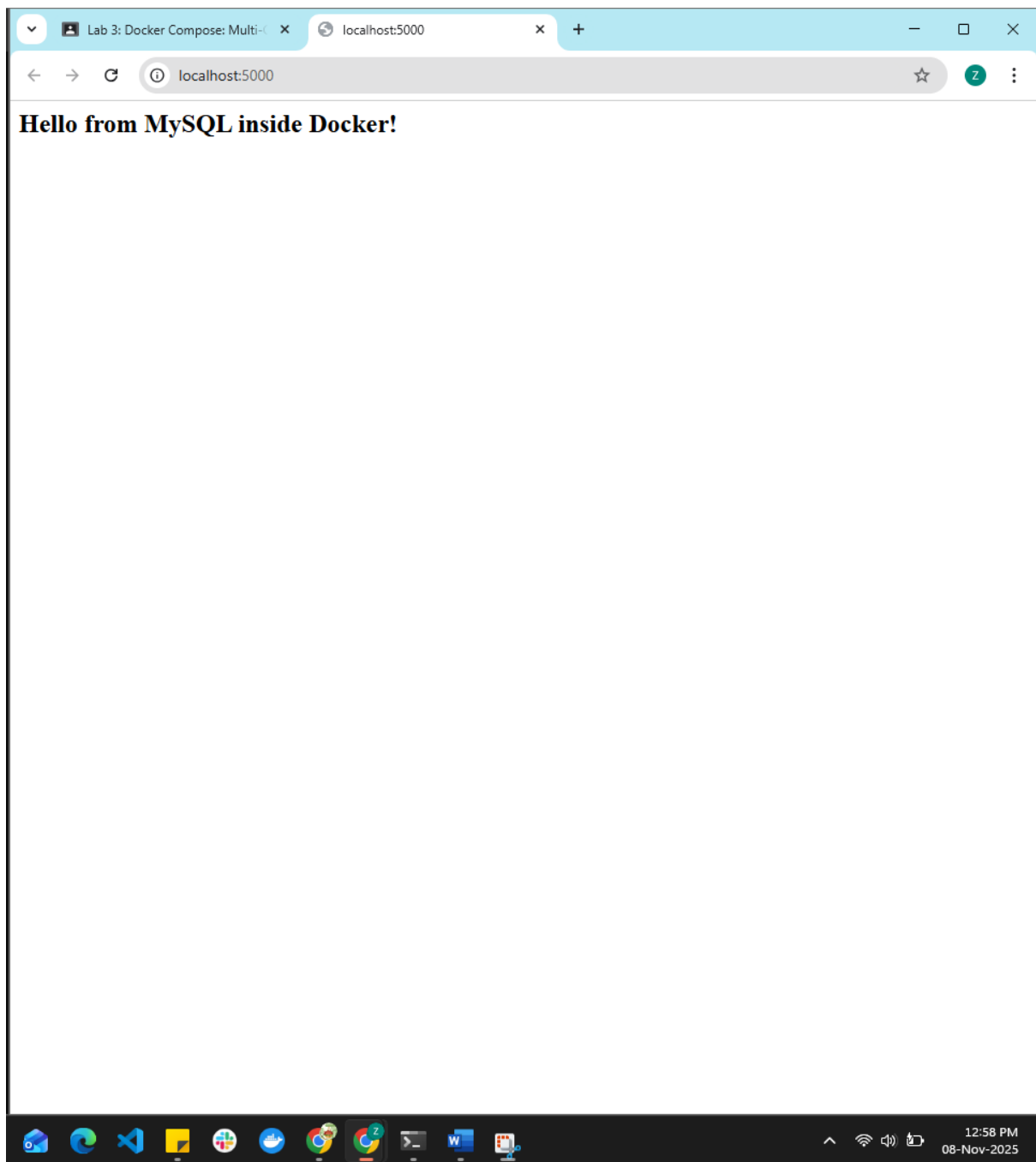
```
version: '3.8'
services:
  web:
    build: .
    ports:
      - "5000:5000"
    depends_on:
      - db
    environment:
      - MYSQL_HOST=db
      - MYSQL_USER=root
      - MYSQL_PASSWORD=root
      - MYSQL_DATABASE=studentdb
  db:
    image: mysql:8.0
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: root
      MYSQL_DATABASE: studentdb
    volumes:
      - db_data:/var/lib/mysql
volumes:
  db_data:
```

The Notepad window also shows a status bar at the bottom: Ln 25, Col 11 | 422 characters | Plain text | 100% | Windows (CRLF) | UTF-8.

**Screenshot #4: Running containers in terminal**

```
zuni_2004@DESKTOP-16K3118: ~/Flask_mysql_app$ docker-compose up --build
WARN[0000] /home/zuni_2004/Flask_mysql_app/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Building 44.2s (13/13) FINISHED
=> [internal] load local image definitions
=> => reading from stdin 510B 0.0s
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 178B 0.0s
=> [internal] load metadata for docker.io/library/python:3.10-slim 2.1s
=> [auth] library/python:pull token for registry-1.docker.io 0.0s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [internal] load build context 0.0s
=> => transferring context: 910B 0.0s
=> [1/5] FROM docker.io/library/python:3.10-slim@sha256:975a1e20a16719060d391eeadac66ee967d70 0.0s
=> => resolve docker.io/library/python:3.10-slim@sha256:975a1e20a16719060d391eeadac66ee967d70 0.0s
=> CACHED [2/5] WORKDIR /app 0.0s
=> [3/5] COPY app/ /app 0.0s
=> [4/5] COPY requirements.txt . 0.1s
=> [5/5] RUN pip install -r requirements.txt 35.0s
=> => exporting layers 2.6s
=> => exporting manifest sha256:2ba1e13c267258eaf6be8257f50c499016fea217529f27f193a61090c35ff7 0.0s
=> => exporting config sha256:6830f8dcf3a348f66f8bc56920f3efe90a2b5a28e15a06d2a297a525ce3c8f3 0.0s
=> => exporting attestation manifest sha256:f61d60a35c702d809432a6d5c2b26c4cfe22658e4d09112e5 0.0s
=> => exporting manifest list sha256:c579f47d4639729a9176801abe31bd9cb10d556c3b038077203e5c353 0.0s
=> => naming to docker.io/library/Flask_mysql_app-web:latest 0.0s
=> => unpacking to docker.io/library/Flask_mysql_app-web:latest 0.0s
=> => resolving provenance for metadata file 0.0s
[+] Running 4/4
 ✓ Flask_mysql_app-web Built 0.0s
 ✓ Network flask_mysql_app_default Created 0.0s
 ✓ Container flask_mysql_app-db-1 Created 0.6s
 ✓ Container flask_mysql_app-web-1 Created 0.1s
Attaching to db-1, web-1
db-1 | 2025-11-08 07:57:06+08:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.44-1.el9 started.
db-1 | 2025-11-08 07:57:06+08:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
db-1 | 2025-11-08 07:57:06+08:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.44-1.el9 started.
db-1 | /var/lib/mysql/mysql.sock' -> '/var/run/mysqld/mysqld.sock'
web-1 | * Serving Flask app 'app'
web-1 | * Debug mode: off
web-1 | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
web-1 | * Running on all addresses (0.0.0.0)
web-1 | * Running on http://127.0.0.1:5000
web-1 | * Running on http://172.15.0.3:5000
web-1 | Press CTRL+C to quit
db-1 | 2025-11-08T07:57:07.2996422 0 [warning] [MY-011068] [Server] The syntax '--skip-host-cache' is deprecated and will be removed in a future release. Please use SET GLOBAL host_cache_size=0 instead.
db-1 | 2025-11-08T07:57:07.3017342 0 [System] [MY-01116] [Server] /usr/sbin/mysqld (mysqld 8.0.44) starting as process 1
db-1 | 2025-11-08T07:57:07.3106877 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
db-1 | 2025-11-08T07:57:07.6075922 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
db-1 | 2025-11-08T07:57:07.8521942 0 [warning] [MY-010968] [Server] CA certificate ca.pem is self signed.
db-1 | 2025-11-08T07:57:07.8522042 0 [System] [MY-013602] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now supported for this channel.
```

Screenshot #5: Browser output



**Screenshot #6: docker ps output**

```
zuni_2004@DESKTOP-16K3I1B X + v
zuni_2004@DESKTOP-16K3I1B:~$ cd flask_mysql_app
zuni_2004@DESKTOP-16K3I1B:~/flask_mysql_app$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
99a731af886b   flask_mysql_app-web                "python app.py"         2 minutes ago Up 2 minutes  0.0.0.0:5
000->5000/tcp, [::]:5000->5000/tcp   flask_mysql_app-web-1    "docker-entrypoint.s..." 2 minutes ago Up 2 minutes  3306/tcp,
6398f490360e   mysql:8.0                          flask_mysql_app-db-1
zuni_2004@DESKTOP-16K3I1B:~/flask_mysql_app$ docker-compose logs web
WARN[0000] /home/zuni_2004/flask_mysql_app/docker-compose.yml: the attribute 'version' is obsolete, it
will be ignored, please remove it to avoid potential confusion
web-1 | * Serving Flask app 'app'
web-1 | * Debug mode: off
web-1 | WARNING: This is a development server. Do not use it in a production deployment. Use a produc
tion WSGI server instead.
web-1 | * Running on all addresses (0.0.0.0)
web-1 | * Running on http://127.0.0.1:5000
web-1 | * Running on http://172.18.0.3:5000
web-1 | Press CTRL+C to quit
web-1 | 172.18.0.1 - - [08/Nov/2025 07:58:27] "GET / HTTP/1.1" 200 -
web-1 | 172.18.0.1 - - [08/Nov/2025 07:58:27] "GET /favicon.ico HTTP/1.1" 404 -
zuni_2004@DESKTOP-16K3I1B:~/flask_mysql_app$ docker-compose logs db
WARN[0000] /home/zuni_2004/flask_mysql_app/docker-compose.yml: the attribute 'version' is obsolete, it
will be ignored, please remove it to avoid potential confusion
db-1 | 2025-11-08 07:57:06+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.44-1.el9
started.
db-1 | 2025-11-08 07:57:06+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
db-1 | 2025-11-08 07:57:06+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.44-1.el9
started.
db-1 | '/var/lib/mysql/mysql.sock' -> '/var/run/mysqld/mysqld.sock'
db-1 | 2025-11-08T07:57:07.299642Z 0 [Warning] [MY-011068] [Server] The syntax '--skip-host-cache' is
deprecated and will be removed in a future release. Please use SET GLOBAL host_cache_size=0 instead.
db-1 | 2025-11-08T07:57:07.301754Z 0 [System] [MY-010116] [Server] /usr/sbin/mysqld (mysqld 8.0.44) s
tarting as process 1
db-1 | 2025-11-08T07:57:07.310657Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
db-1 | 2025-11-08T07:57:07.607592Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
db-1 | 2025-11-08T07:57:07.852194Z 0 [Warning] [MY-010068] [Server] CA certificate ca.pem is self sig
ned.
db-1 | 2025-11-08T07:57:07.852244Z 0 [System] [MY-013602] [Server] Channel mysql_main configured to s
upport TLS. Encrypted connections are now supported for this channel.
db-1 | 2025-11-08T07:57:07.856451Z 0 [Warning] [MY-011810] [Server] Insecure configuration for --pid-
file: Location '/var/run/mysqld' in the path is accessible to all OS users. Consider choosing a differ
ent directory.
db-1 | 2025-11-08T07:57:07.880056Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Bi
nd-address: '::' port: 33060, socket: /var/run/mysqld/mysqld.sock
db-1 | 2025-11-08T07:57:07.880341Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connec
tions. Version: '8.0.44' socket: '/var/run/mysqld/mysqld.sock' port: 3306 MySQL Community Server -
GPL.
zuni_2004@DESKTOP-16K3I1B:~/flask_mysql_app$ |
```

Screenshot #7: docker-compose down output

```
zuni_2004@DESKTOP-16K3I1B:~/flask_mysql_app$ docker-compose down
WARN[0000] /home/zuni_2004/flask_mysql_app/docker-compose.yml: the attribute 'version' is obsolete, it
will be ignored, please remove it to avoid potential confusion
[+] Running 3/3
✔Container flask_mysql_app-web-1   Removed      1.6s
✔Container flask_mysql_app-db-1    Removed      1.5s
✔Network flask_mysql_app_default   Removed      0.7s
zuni_2004@DESKTOP-16K3I1B:~/flask_mysql_app$

zuni_2004@DESKTOP-16K3I1B:~/flask_mysql_app$ docker-compose down -v
docker image prune -f
WARN[0000] /home/zuni_2004/flask_mysql_app/docker-compose.yml: the attribute 'version' is obsolete, it
will be ignored, please remove it to avoid potential confusion
[+] Running 1/1
✔Volume flask_mysql_app_db_data    Removed      0.0s
Total reclaimed space: 0B
zuni_2004@DESKTOP-16K3I1B:~/flask_mysql_app$
```

## Observations

1. **Docker Compose is much easier** than running containers one by one. A single command (docker-compose up) can start, link, and network an entire application.
2. **Containers can talk to each other by name.** The Flask app was able to find the database by using the hostname db, which was the name of the service in the docker-compose.yml file.
3. **Data is safe with volumes.** Even after stopping and removing the database container with docker-compose down, the db\_data volume protected the data, so it was still there the next time we started the app.

## Conclusion

This lab was a great introduction to Docker Compose. I learned how to use a single docker-compose.yml file to define and run a complete application with multiple services. This makes it simple to manage a web app and a database together. The most important things I learned were how to connect containers (networking) and how to save data (volumes).